DataWareHouse and DataMinning

Lab Experiments

EXPERIMENT-1

ADDITION:

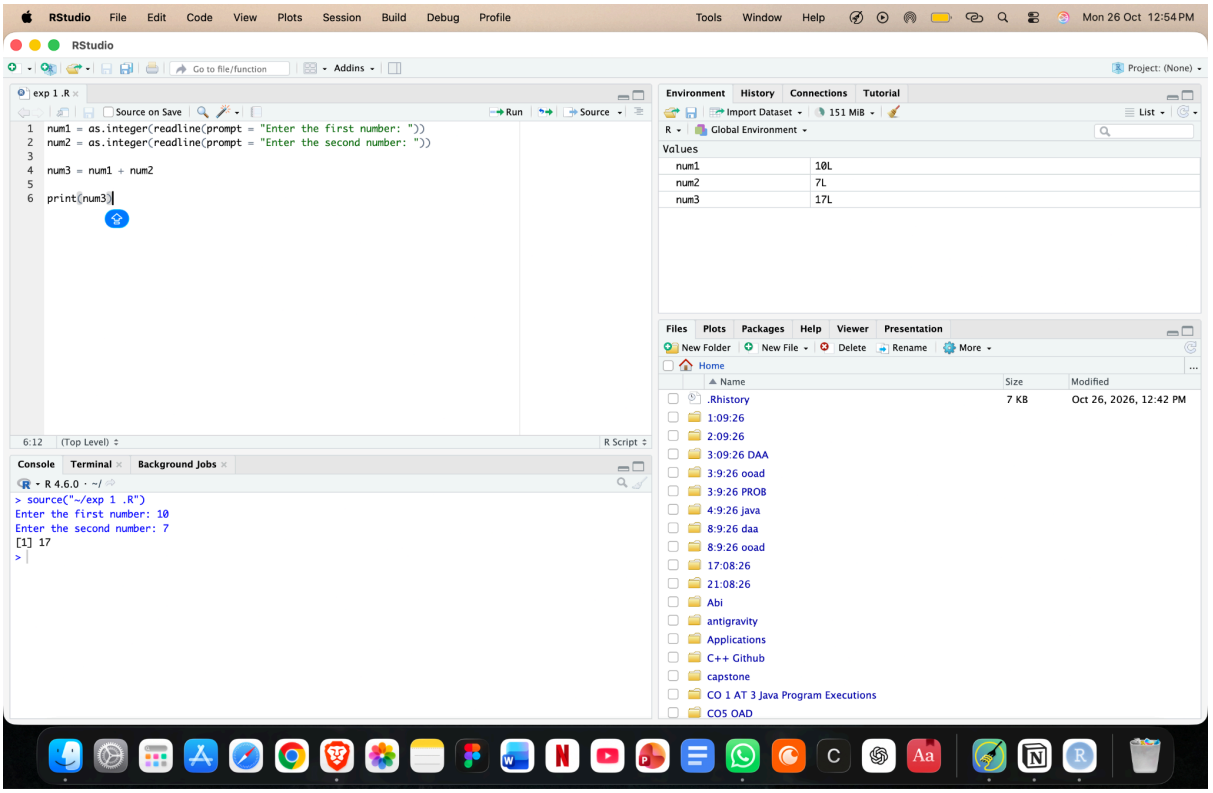
AIM:

To prove the program for addition using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1+num2
print(num3)
```

INPUT & OUTPUT :



RESULT:

Thus the basic program addition is executed successfully.

EXPERIMENT-2

SUBTRACTION:

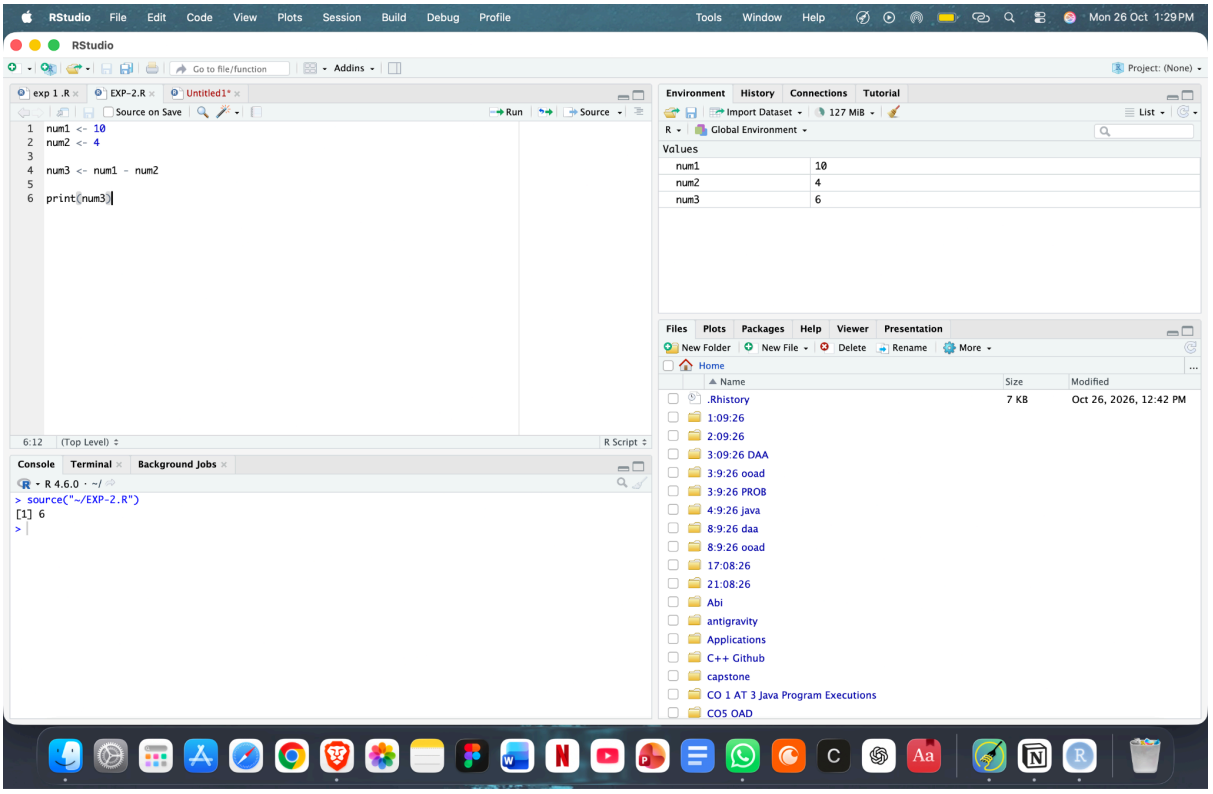
AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

```
num1 <- 10
num2 <- 4
num3 <- num1 - num2
print(num3)
```

OUTPUT:



RESULT:

Thus, the basic program subtraction is executed successfully.

EXPERIMENT-3

MULTIPLICATION:

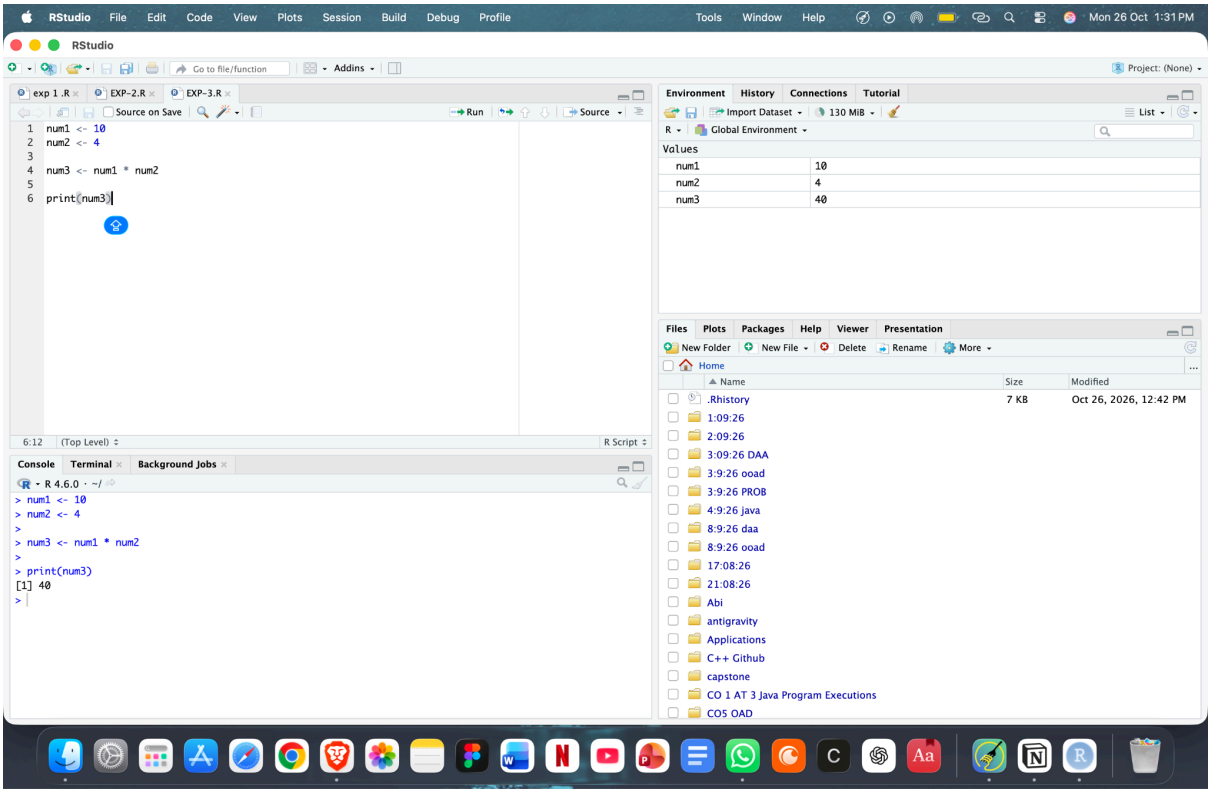
AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

```
num1 <- 10
num2 <- 4
num3 <- num1 * num2
print(num3)
```

OUTPUT:



RESULT:

Thus, the basic program multiplication is executed successfully.

EXPERIMENT-4

DIVISION:

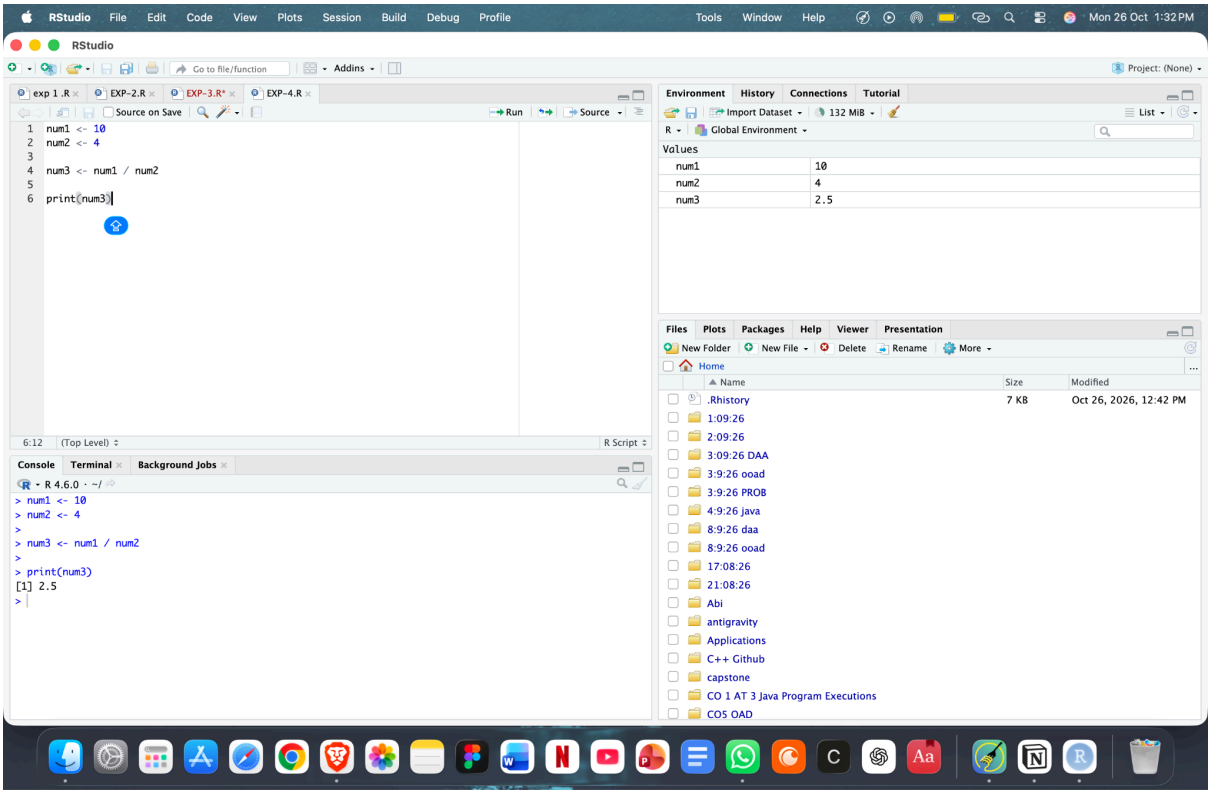
AIM:

To prove the program for division using R-tool.

PROGRAM:

```
num1 <- 10
num2 <- 4
num3 <- num1 /  num2
print(num3)
```

OUTPUT:



RESULT:

Thus, the basic program division was executed successfully.

EXPERIMENT-5

ODD OR EVEN:

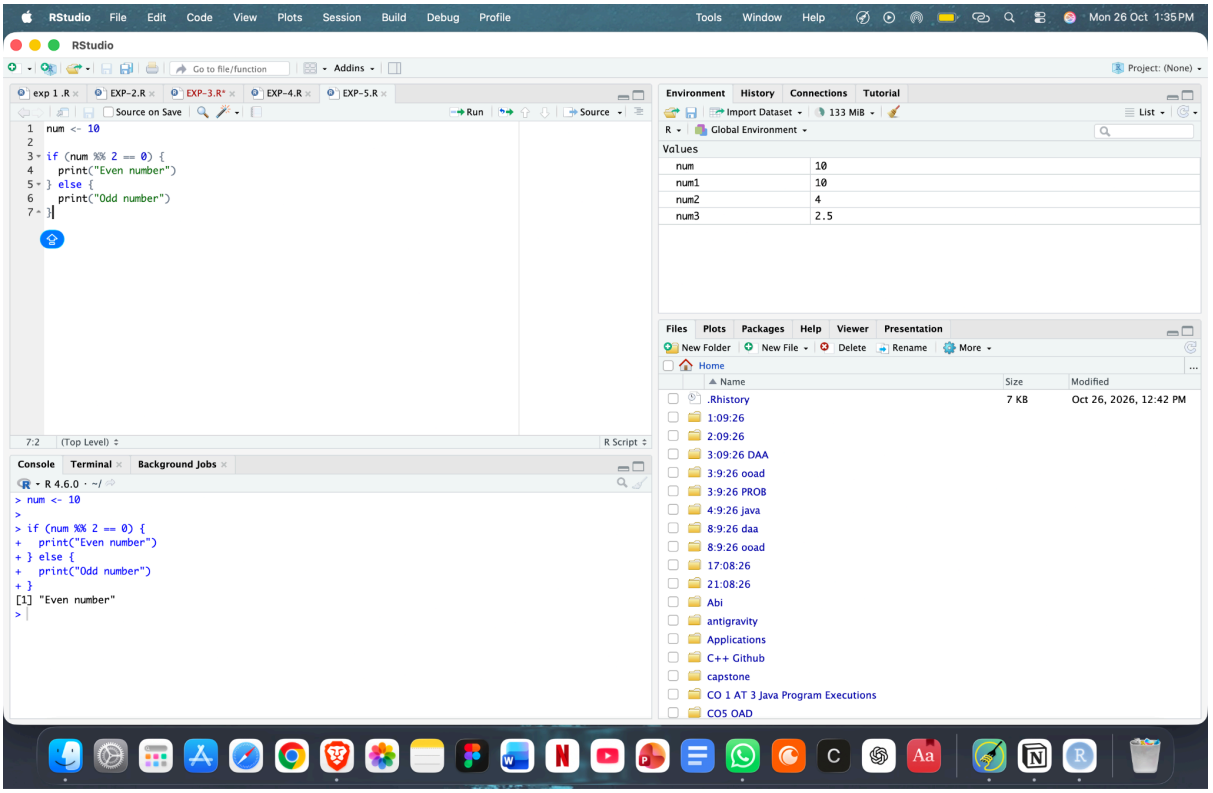
AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
num <- 10
if (num %% 2 == 0) {
  print("Even number")
} else {
  print("Odd number")
}
```

OUTPUT:



RESULT:

Thus, the basic program odd or even was executed successfully.

EXPERIMENT-6

MEAN,MEDIAN,MODE:

AIM:

To write the program for mean,median,mode.

PROGRAM:

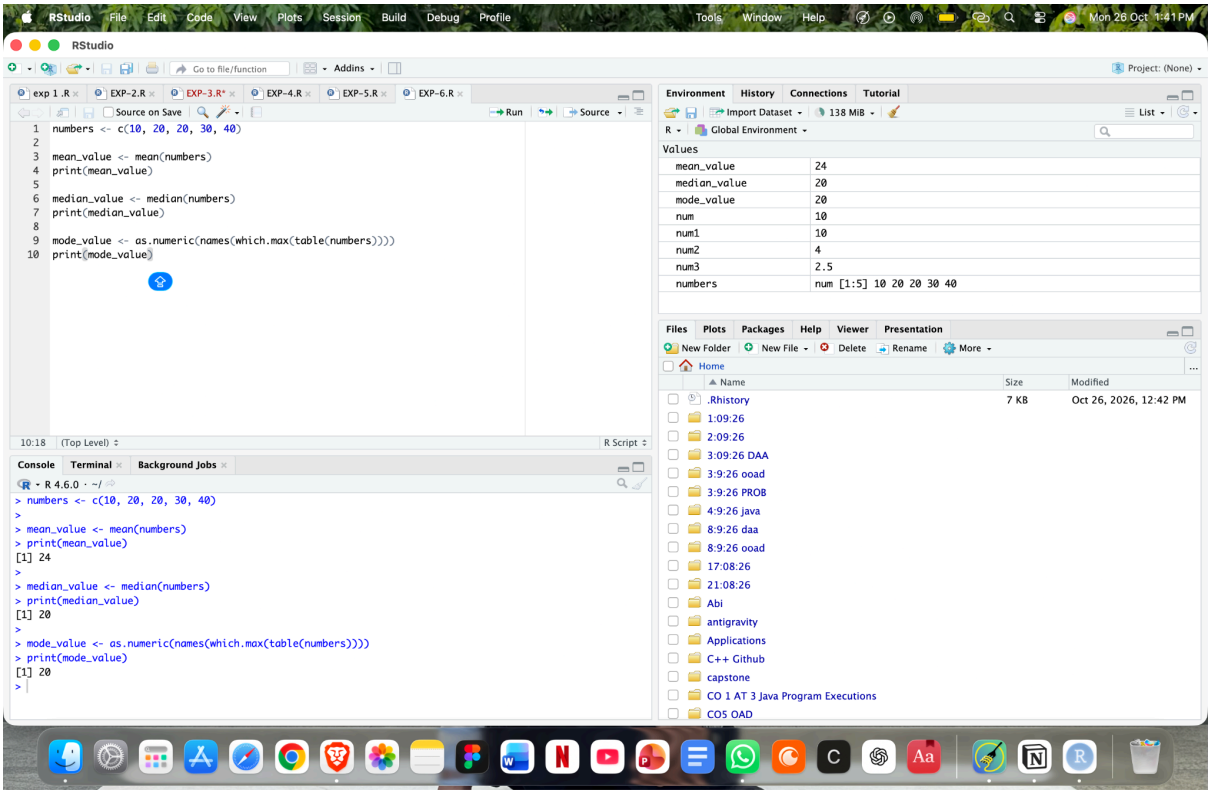
```
numbers <- c(10, 20, 20, 30, 40)

mean_value <- mean(numbers)
print(mean_value)

median_value <- median(numbers)
print(median_value)

mode_value <- as.numeric(names(which.max(table(numbers))))
print(mode_value)
```

OUTPUT:



RESULT:

Thus the central tendency and measure of dispersion is executed successfully.

EXPERIMENT-7

SUMMARY:

AIM:

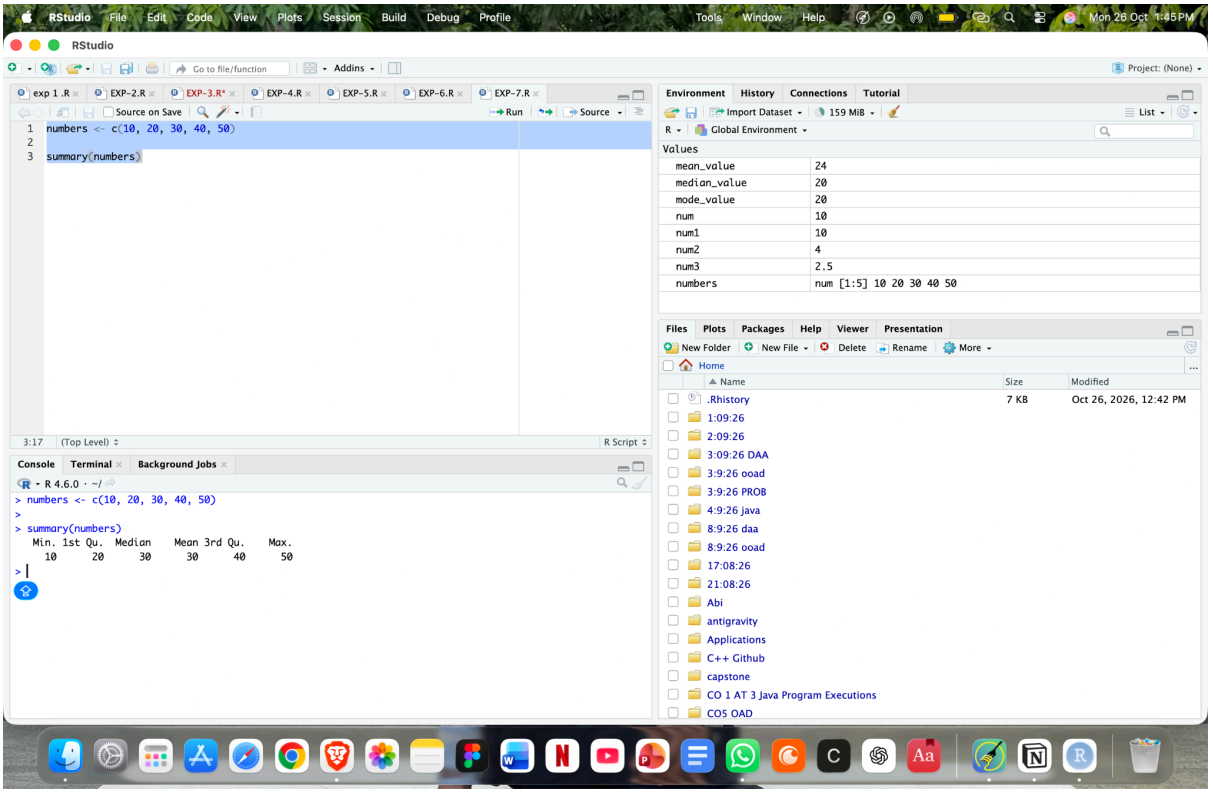
To write the program for summary using R-tool.

PROGRAM:

numbers <- c(10, 20, 30, 40, 50)

summary(numbers)

OUTPUT:



RESULT:

Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT-8

GREATER AMONG THREE NUMBERS:

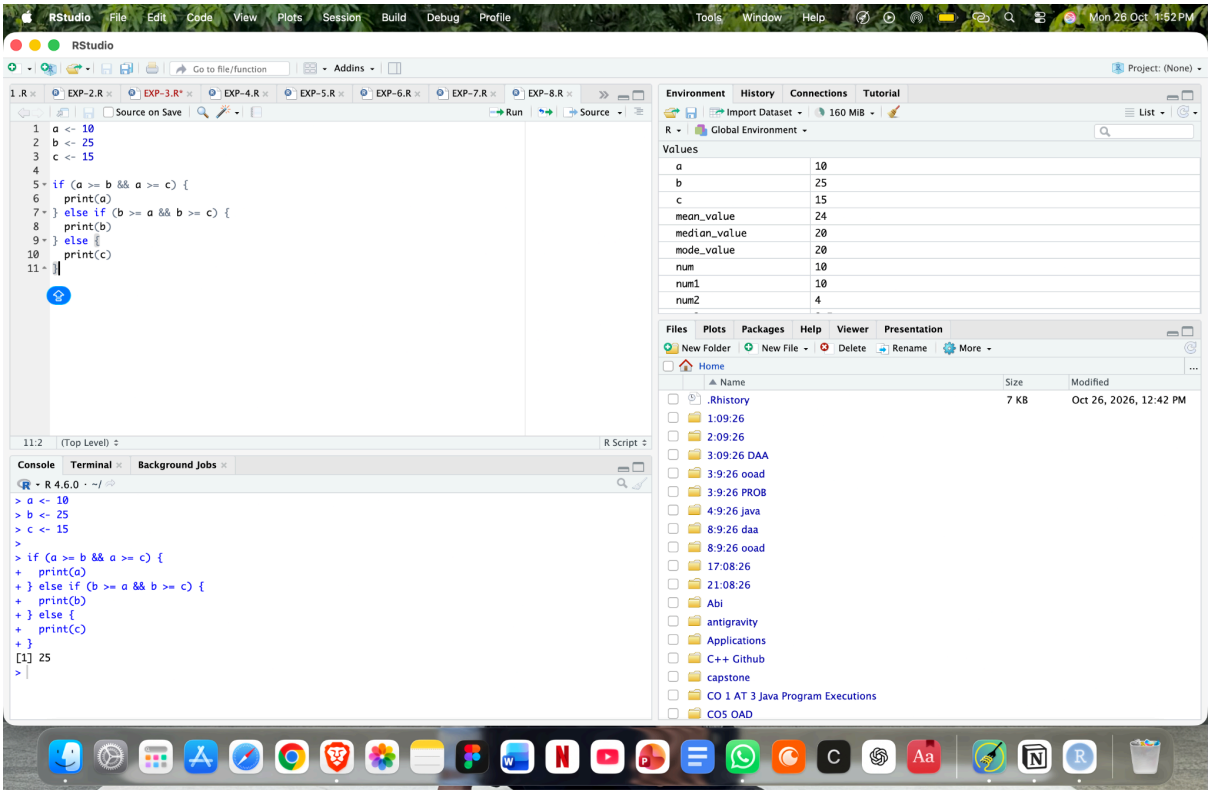
AIM:

To write the program for the greatest among three numbers.

PROGRAM:

```
a <- 10
b <- 25
c <- 15
if (a >= b && a >= c) {
  print(a)
} else if (b >= a && b >= c) {
  print(b)
} else {
  print(c)
}
```

OUTPUT:



RESULT:

Thus, the greatest among the three numbers was executed successfully.

EXPERIMENT-9

IQR:

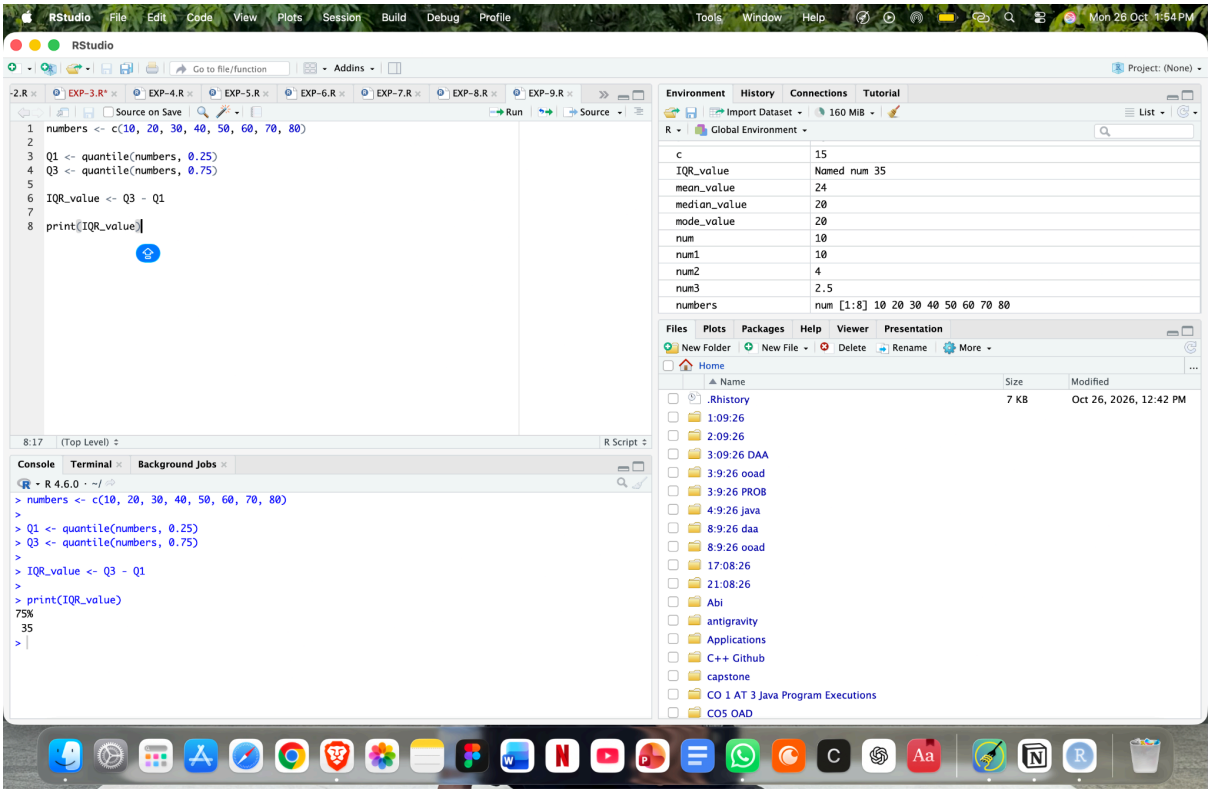
AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
numbers <- c(10, 20, 30, 40, 50, 60, 70, 80)
Q1 <- quantile(numbers, 0.25)
Q3 <- quantile(numbers, 0.75)
IQR_value <- Q3 - Q1
print(IQR_value)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10

QUANTILE:

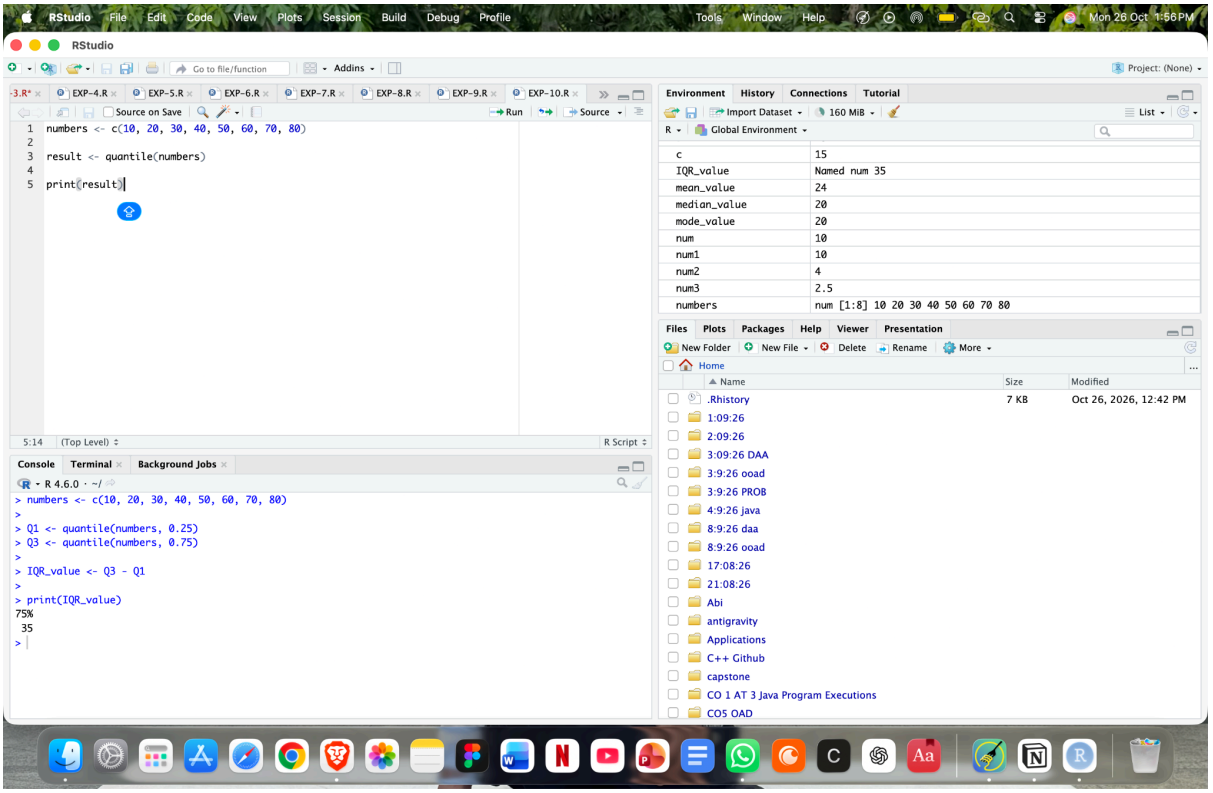
AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
numbers <- c(10, 20, 30, 40, 50, 60, 70, 80)
result <- quantile(numbers)
print(result)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11

MID RANGE:

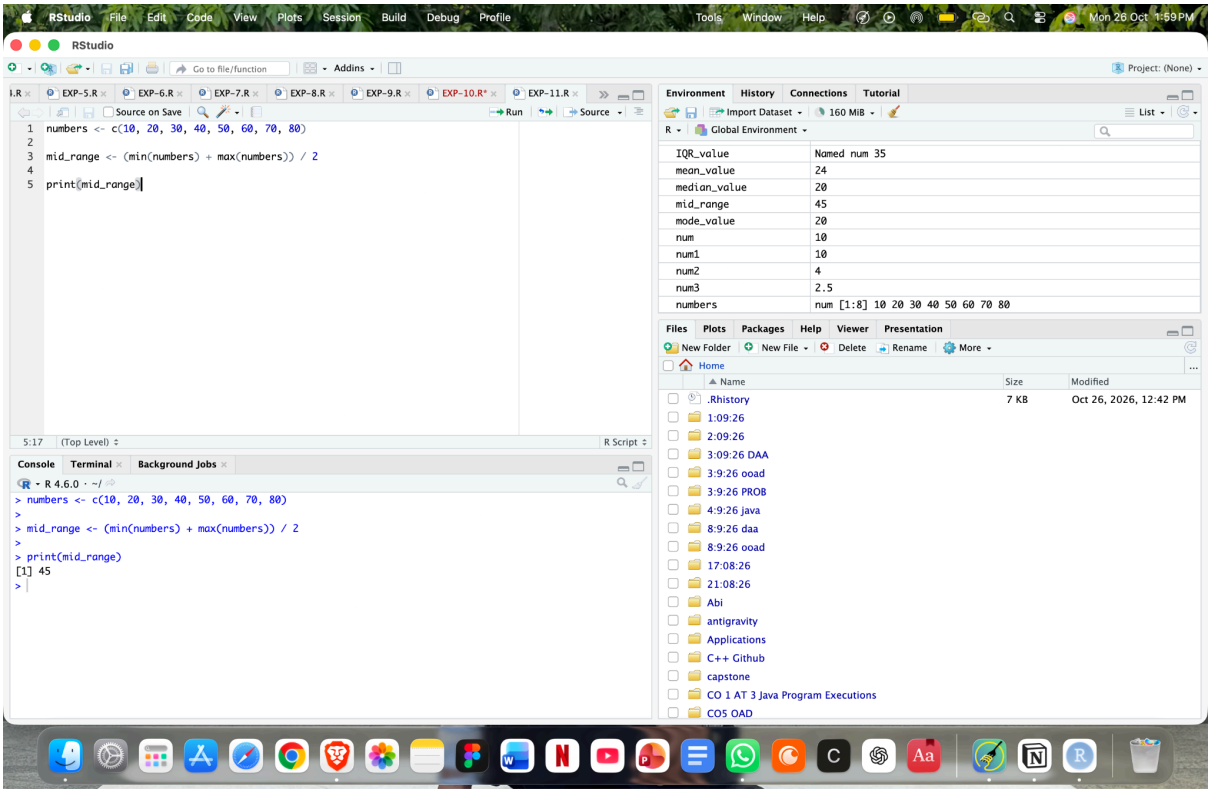
AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
numbers <- c(10, 20, 30, 40, 50, 60, 70, 80)
mid_range <- (min(numbers) + max(numbers)) / 2
print(mid_range)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-12

Z-SCOORE NORMALIZATION:

AIM:

To write the program for Z-scoore normalization using R-tool.

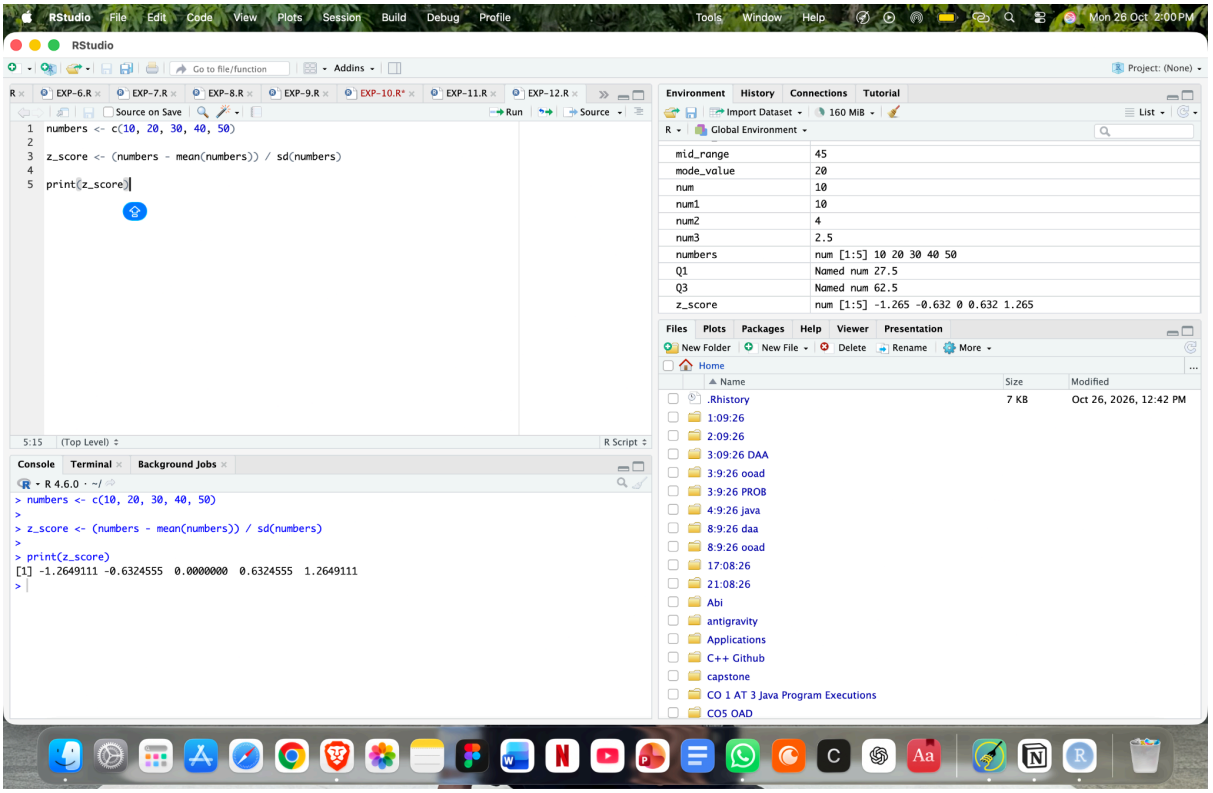
PROGRAM:

```
numbers <- c(10, 20, 30, 40, 50)

z_score <- (numbers - mean(numbers)) / sd(numbers)

print(z_score)
```

OUTPUT:



RESULT:
Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT-13
MIN,MAX,MEAN,MINMAX:
AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

```
numbers <- c(10, 20, 30, 40, 50)


print(min(numbers))

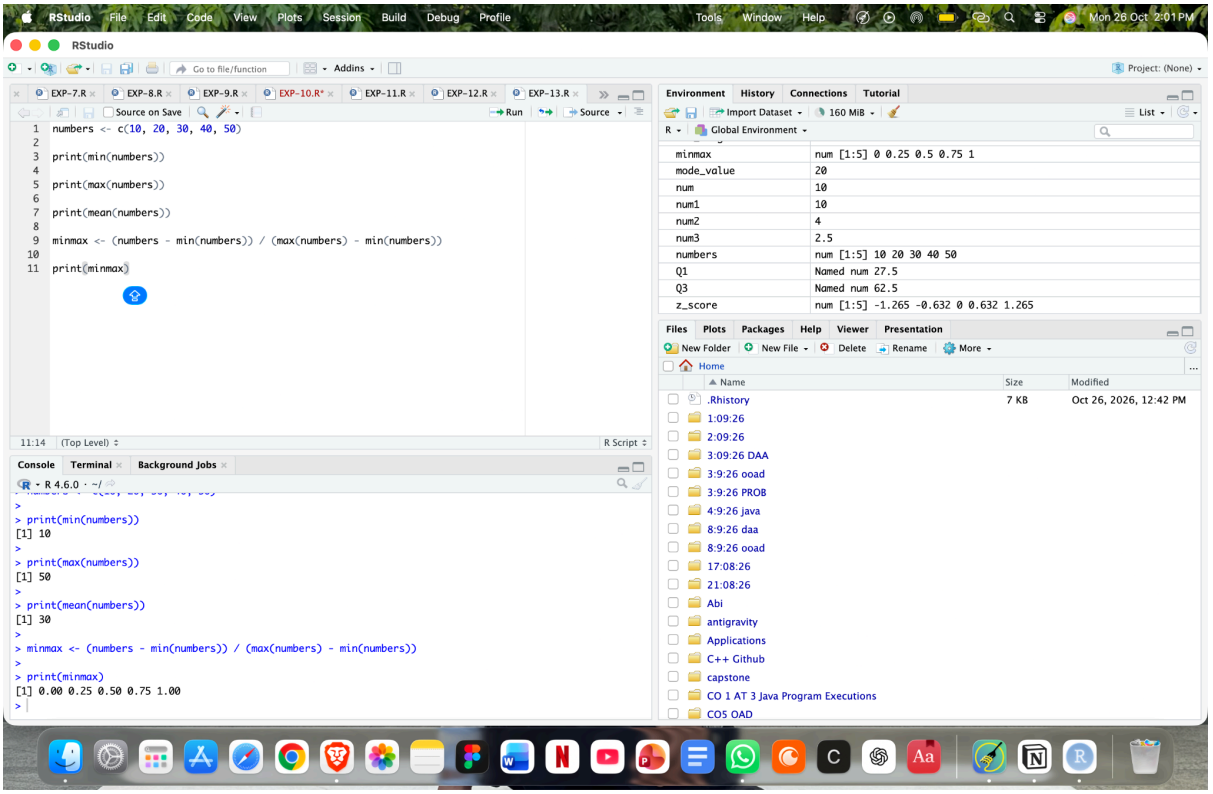

print(max(numbers))


print(mean(numbers))


minmax <- (numbers - min(numbers)) / (max(numbers) - min(numbers))


print(minmax)
```

OUTPUT:



RESULT:

Thus, the program for min,max,minmax,mean was executed successfully.

EXPERIMENT-14

BAR PLOT AND HORIZONTAL BAR:

AIM:

marks <- c(80, 70, 90, 60, 85)

subjects <- c("Tamil", "English", "Maths", "Science", "Computer")

Vertical Bar Plot

barplot(marks,

names.arg = subjects,

main = "Subject-wise Marks",

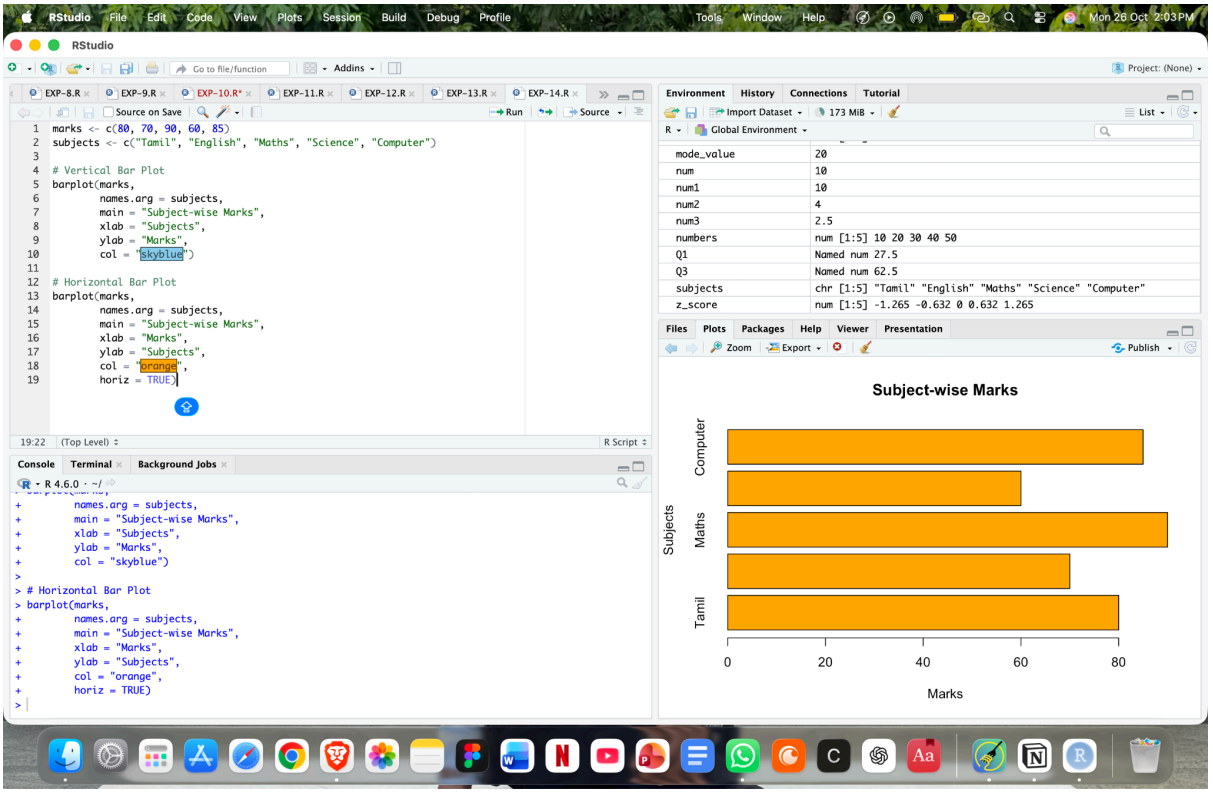
xlab = "Subjects",

ylab = "Marks",

col = "skyblue")

```
# Horizontal Bar Plot
barplot(marks,
        names.arg = subjects,
        main = "Subject-wise Marks",
        xlab = "Subjects",
        ylab = "Marks",
        col = "orange",
        horiz = TRUE)
```

OUTPUT:



RESULT:

Thus, the bar and horizontal bar plot was executed successfully.

EXPERIMENT-15

BOX PLOT:

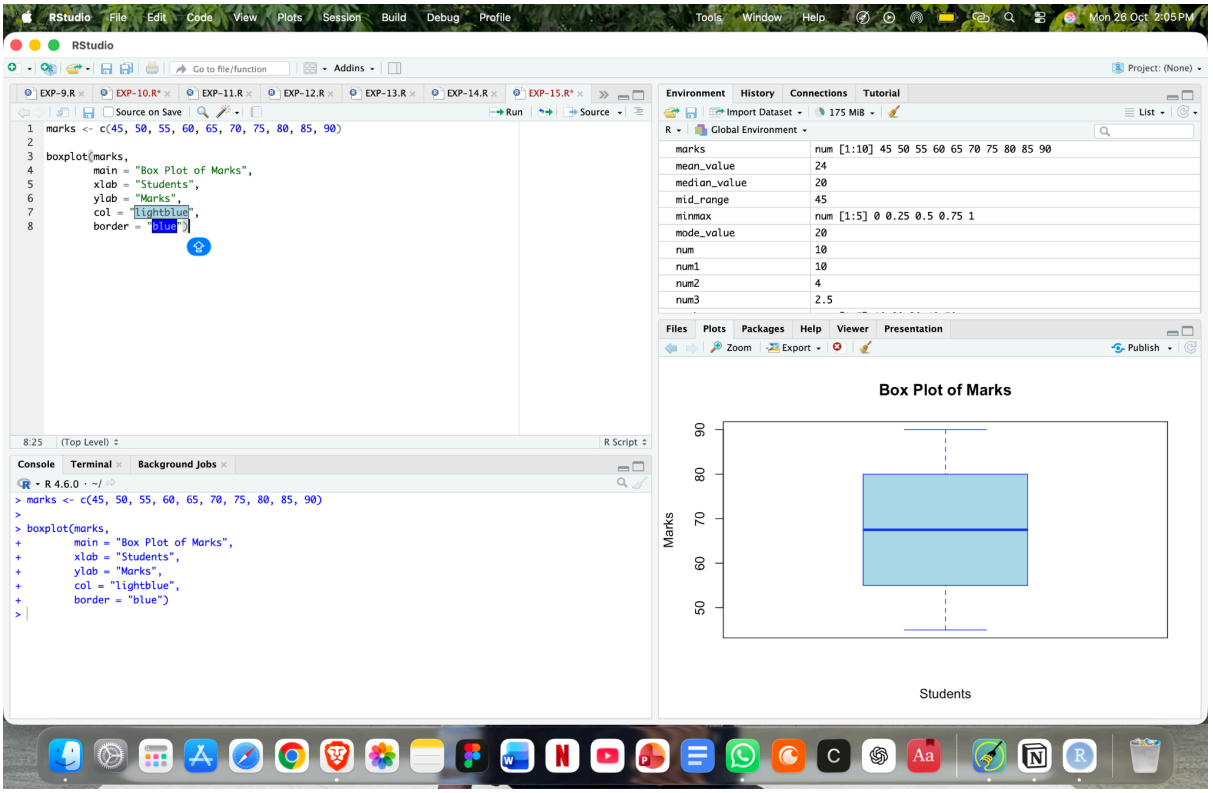
AIM:

To draw the box plot using R-tool.

PROGRAM:

```
marks <- c(45, 50, 55, 60, 65, 70, 75, 80, 85, 90)
boxplot(marks,
        main = "Box Plot of Marks",
        xlab = "Students",
        ylab = "Marks",
        col = "lightblue",
        border = "blue")
```

OUTPUT:



RESULT:

Thus, the box plot was executed successfully.

EXPERIMENT-16

HISTOGRAM:

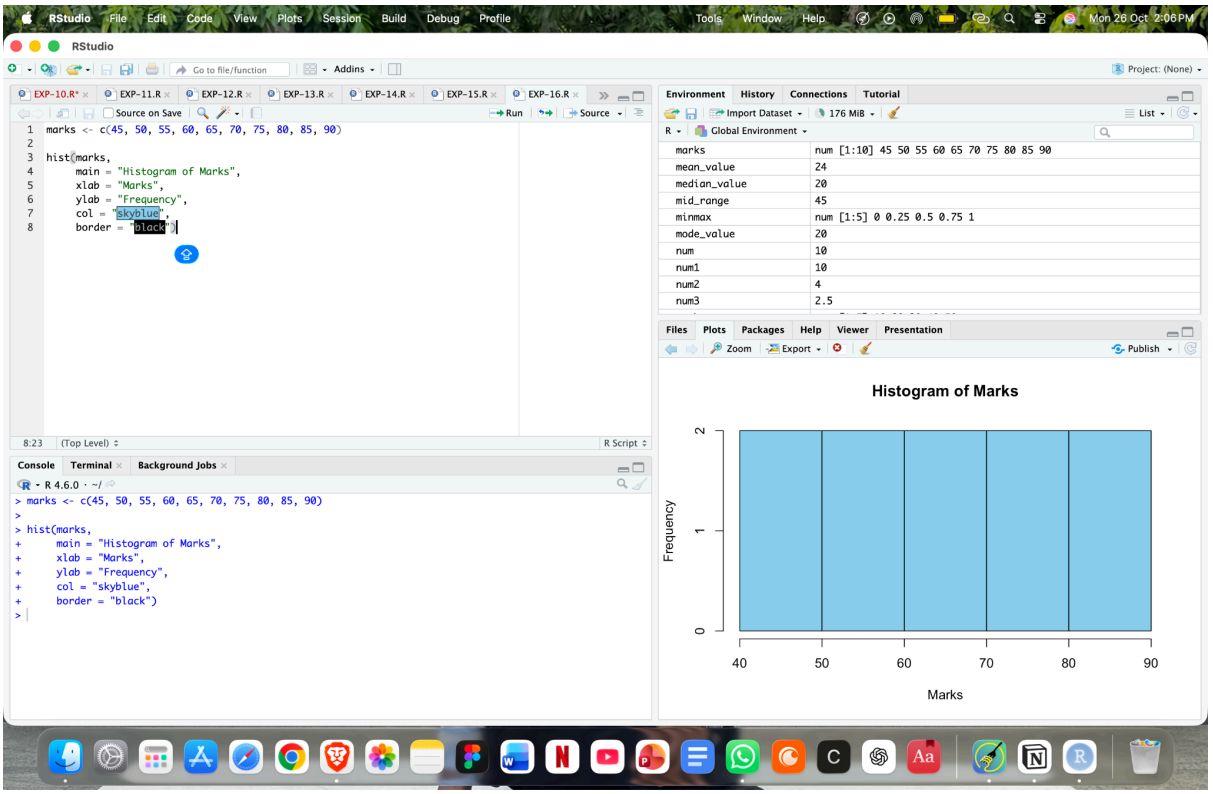
AIM:

To draw the histogram plot using R-tooll.

PROGRAM:

```
marks <- c(45, 50, 55, 60, 65, 70, 75, 80, 85, 90)
hist(marks,
     main = "Histogram of Marks",
     xlab = "Marks",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")
```

OUTPUT:



RESULT:

Thus, the histogram plot was executed successfully.

EXPERIMENT-17

CORRELATION ANALYSIS:

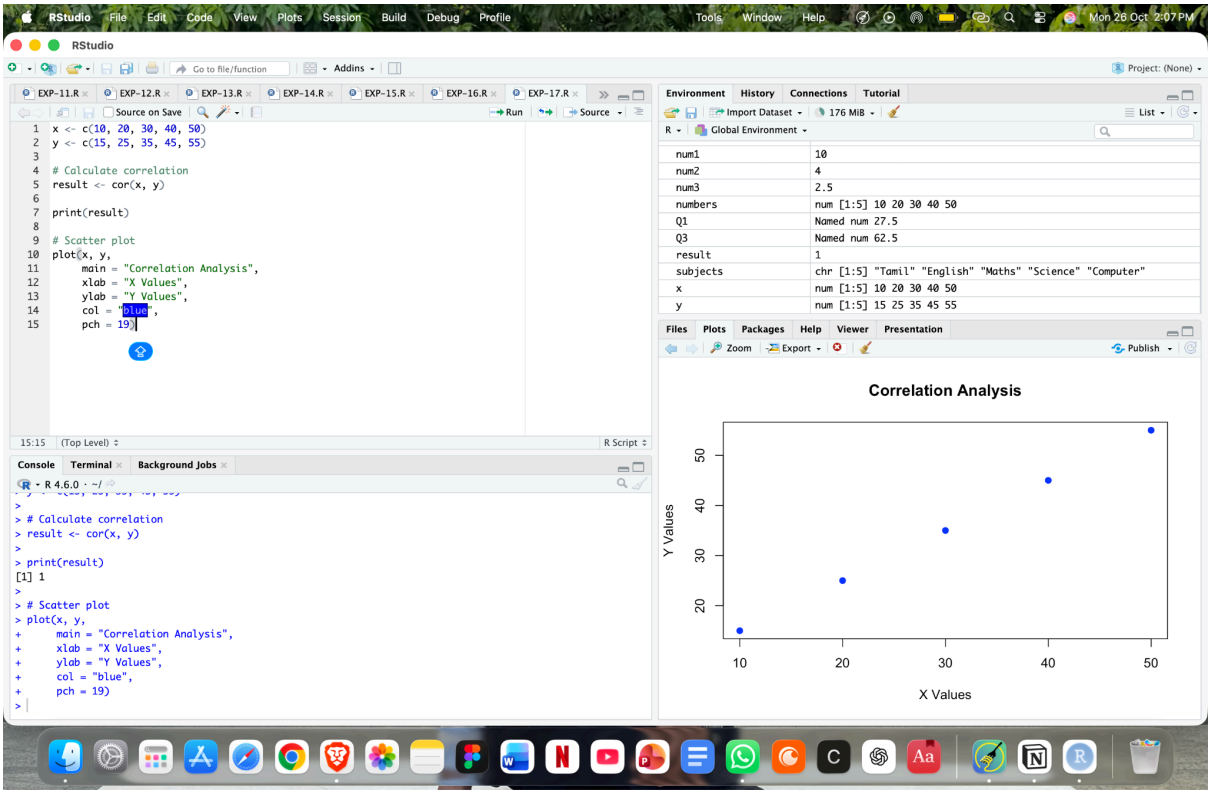
AIM:

To write the program for correlation analysis using R-tool.

PROGRAM:

```
x <- c(10, 20, 30, 40, 50)
y <- c(15, 25, 35, 45, 55)
# Calculate correlation
result <- cor(x, y)
print(result)
# Scatter plot
plot(x, y,
     main = "Correlation Analysis",
     xlab = "X Values",
     ylab = "Y Values",
     col = "blue",
     pch = 19)
```

OUTPUT:



RESULT:

Thus, the correlation analysis was executed successfully.

EXPERIMENT-18

SCATTER PLOT:

AIM:

To draw the scatter plot using R-tool

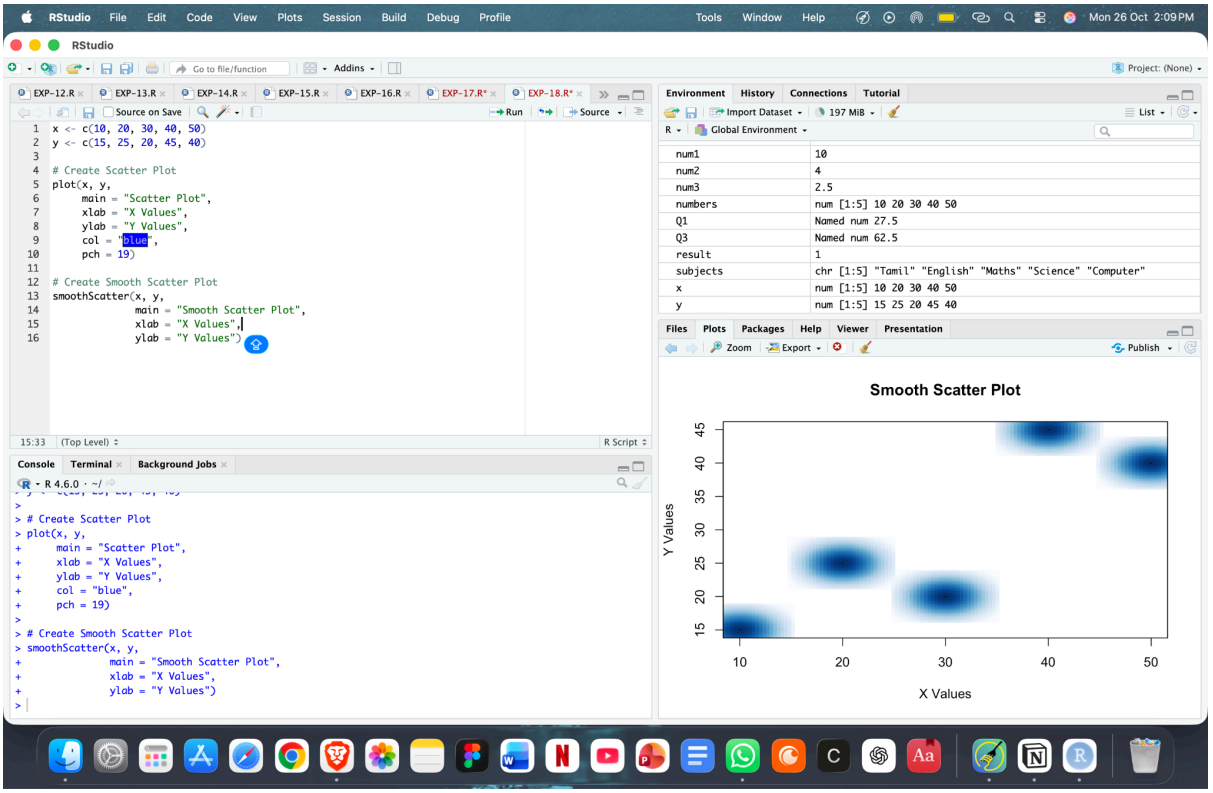
PROGRAM:

```
x <- c(10, 20, 30, 40, 50)
y <- c(15, 25, 20, 45, 40)

# Create Scatter Plot
plot(x, y,
    main = "Scatter Plot",
    xlab = "X Values",
    ylab = "Y Values",
    col = "blue",
    pch = 19)

# Create Smooth Scatter Plot
smoothScatter(x, y,
            main = "Smooth Scatter Plot",
            xlab = "X Values",
            ylab = "Y Values")
```

OUTPUT:



RESULT:

Thus, the scatter plot was executed successfully.

EXPERIMENT-19

LINEAR REGRESSION:

AIM:

To write thr program for the linear regression using R-tool.

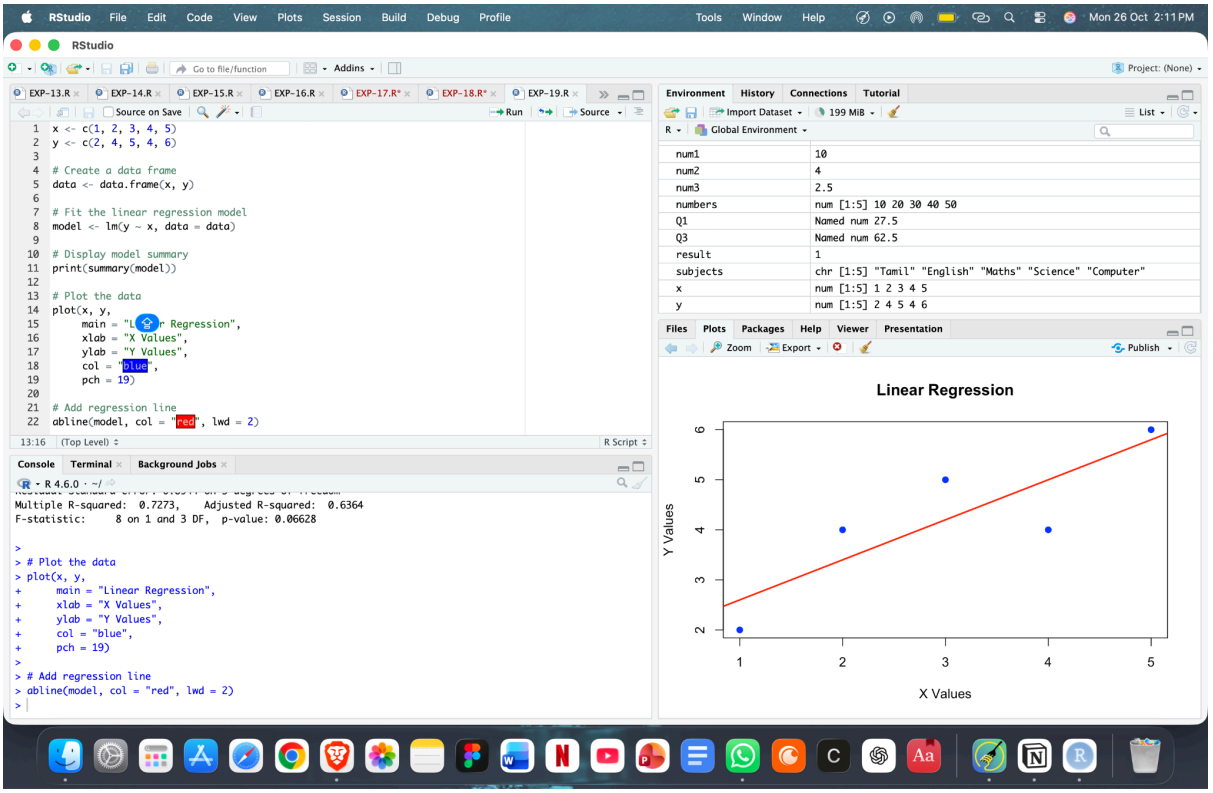
PROGRAM:

```
x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 5, 4, 6)
# Create a data frame
data <- data.frame(x, y)
# Fit the linear regression model
model <- lm(y ~ x, data = data)
# Display model summary
print(summary(model))

# Plot the data
```

```
plot(x, y,
     main = "Linear Regression",
     xlab = "X Values",
     ylab = "Y Values",
     col = "blue",
     pch = 19)
# Add regression line
abline(model, col = "red", lwd = 2)
```

OUTPUT:



RESULT:

Thus, the linear regression program was executed successfully.

EXPERIMENT-20

MULTIPLE REGRESSION:

AIM:

To write the program for the multiple regression.

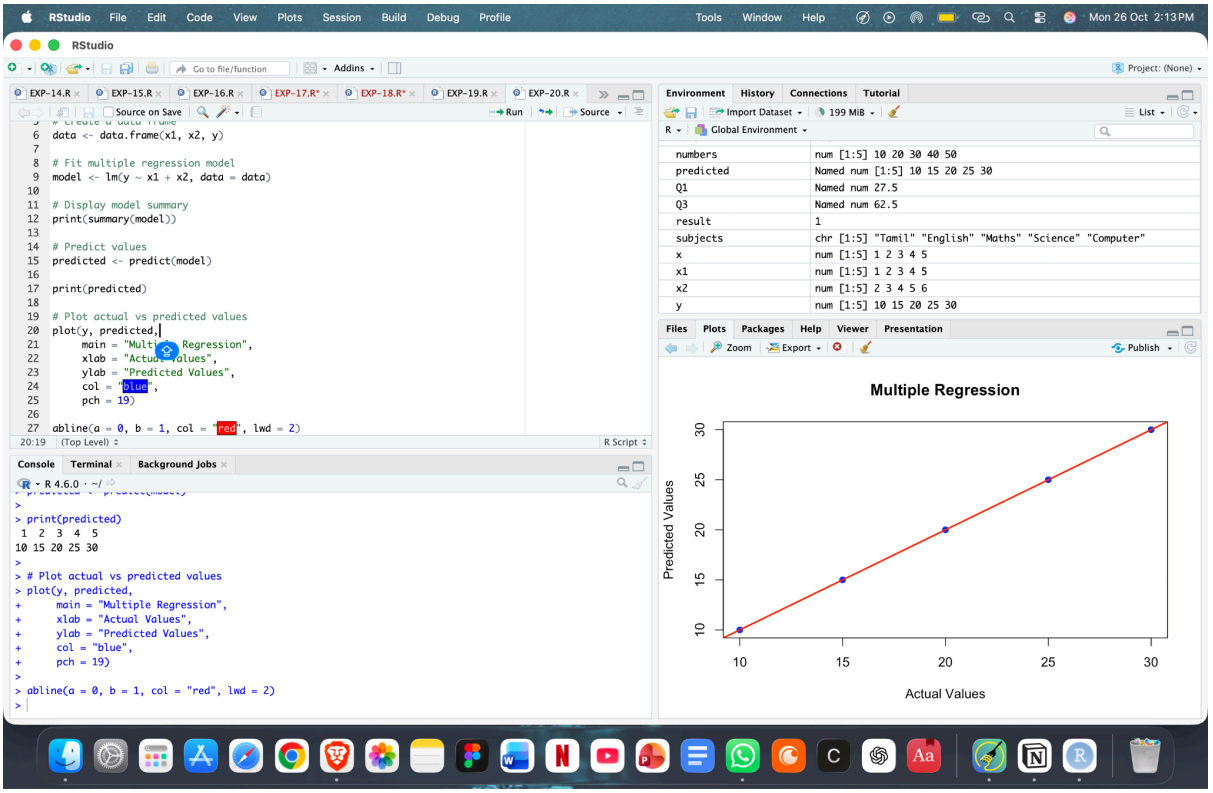
PROGRAM:

```
x1 <- c(1, 2, 3, 4, 5)
```

```
x2 <- c(2, 3, 4, 5, 6)
y <- c(10, 15, 20, 25, 30)

# Create a data frame
data <- data.frame(x1, x2, y)
# Fit multiple regression model
model <- lm(y ~ x1 + x2, data = data)
# Display model summary
print(summary(model))
# Predict values
predicted <- predict(model)
print(predicted)
# Plot actual vs predicted values
plot(y, predicted,
     main = "Multiple Regression",
     xlab = "Actual Values",
     ylab = "Predicted Values",
     col = "blue",
     pch = 19)
abline(a = 0, b = 1, col = "red", lwd = 2)
```

OUTPUT:



RESULT:

Thus, the Multiple regression program was executed successfully.